

SUPPORTING VULNERABLE HHs TO RECONSTRUCT

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Dhokaima Café, PATAN DHOKA

PROJECT TITLE:

**RESILIENT COMMUNITIES THROUGH BUILDING BACK
BETTER IN DISTRICTS MOST SEVERELY AFFECTED BY 2015
EARTHQUAKE**

DURATION:

MAY 2016 TO DECEMBER 2017 (20 MONTHS)

IMPLEMENTATION

PARTNERS:

**NATIONAL RECONSTRUCTION AUTHORITY,
DEPARTMENT OF URBAN DEVELOPMENT AND
BUILDING CONSTRUCTION, GAUPALIKA_s,
NAGARPALIKA_s, LOCAL NGOs**



SCALE OF SUPPORT

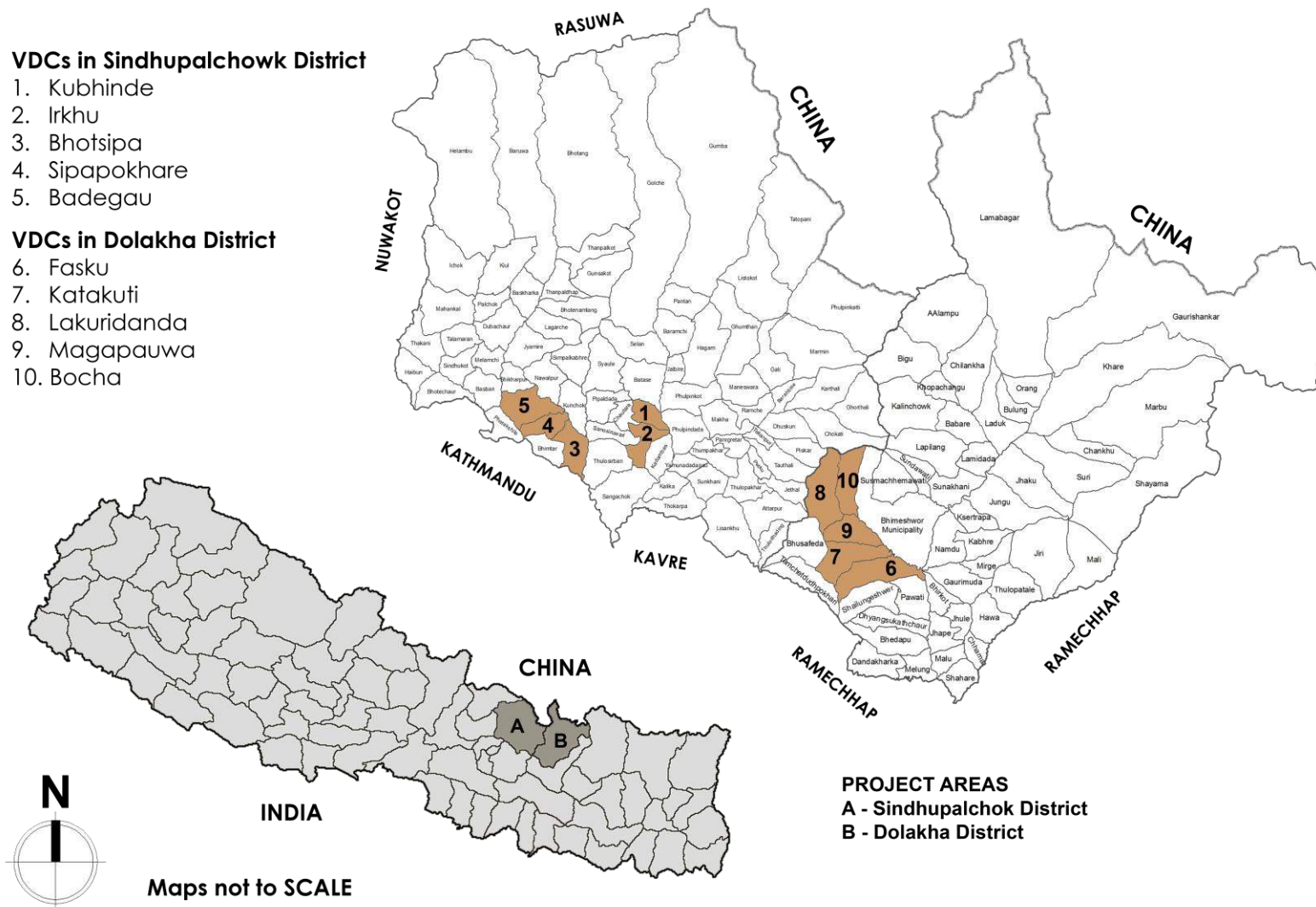
- 15 Wards
10 erstwhile VDCs
- Total Population: 55,260

VDCs in Sindhupalchowk District

1. Kubhinde
2. Irkhu
3. Bhotsipa
4. Sipapokhare
5. Badegau

VDCs in Dolakha District

6. Fasku
7. Katakuti
8. Lakuridanda
9. Magapauwa
10. Bocha



APPROACH & IMPLEMENTATION STRATEGY

SOCIO-TECHNICAL Facilitation Process undertaken to equip the communities, particularly the most-at-risk HHs affected by the earthquake, with a reconstruction action plan and building necessary capacities through information, knowledge and skills to undertake owner-driven reconstruction of disaster resistant housing leading to building their resilience to future risks.

Key Elements of socio-technical facilitation process

1. **Reconstruction Action Plan**
2. **On-the-job training (OJT)** through construction of Technology Demonstration Houses (TDHs)
3. Door to door socio-technical advisory services by **Awaz Nirman Sathi (ANS)**
4. **Safer Construction Clinic**
5. **Reconstruction Grant**
6. **Resilience Fund**

B. APPROACH & IMPLEMENTATION STRATEGY

RECONSTRUCTION ACTION PLAN (RAP)

- **RAP** - the guiding document at ward level to plan and undertake inclusive reconstruction. An inclusive process ensured with participation from the most-at-risk HHs in hazard, vulnerability and capacity assessment, reconstruction needs, house design and challenges in the process.
- **Documents (both written and through pictorial representation) - issues hindering reconstruction, particularly on**
 - a) Land b) Materials c) Labor d) Finance e) Technology
- **Focuses on**
 - a) what are the pertinent issues/ need b) what are the available resources
 - c) what are the constraints, process of prioritization and phasing
 - d) who are responsible and identification of potential sources of support



B. APPROACH & IMPLEMENTATION STRATEGY

RECONSTRUCTION ACTION PLAN (RAP) [RESULTS]



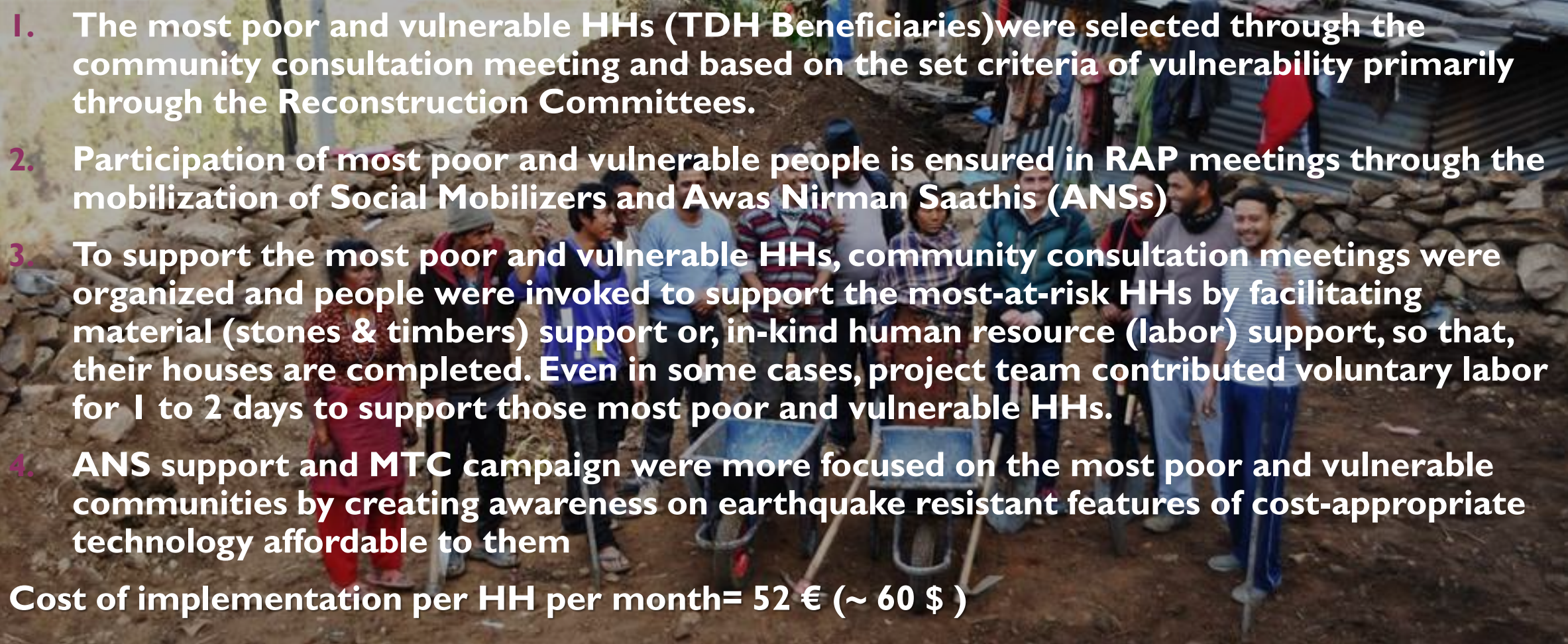
Reconstruction Committee meeting



VDC level consultation meeting

10 VDCs undertook reconstruction in accordance with the **Reconstruction Action Plan (RAP)** through an inclusive process (representation of **vulnerable communities in Reconstruction Committees**), **Mobilization of support for reconstruction of most-vulnerable HHs** through Community Mobilization and involvement of local representatives

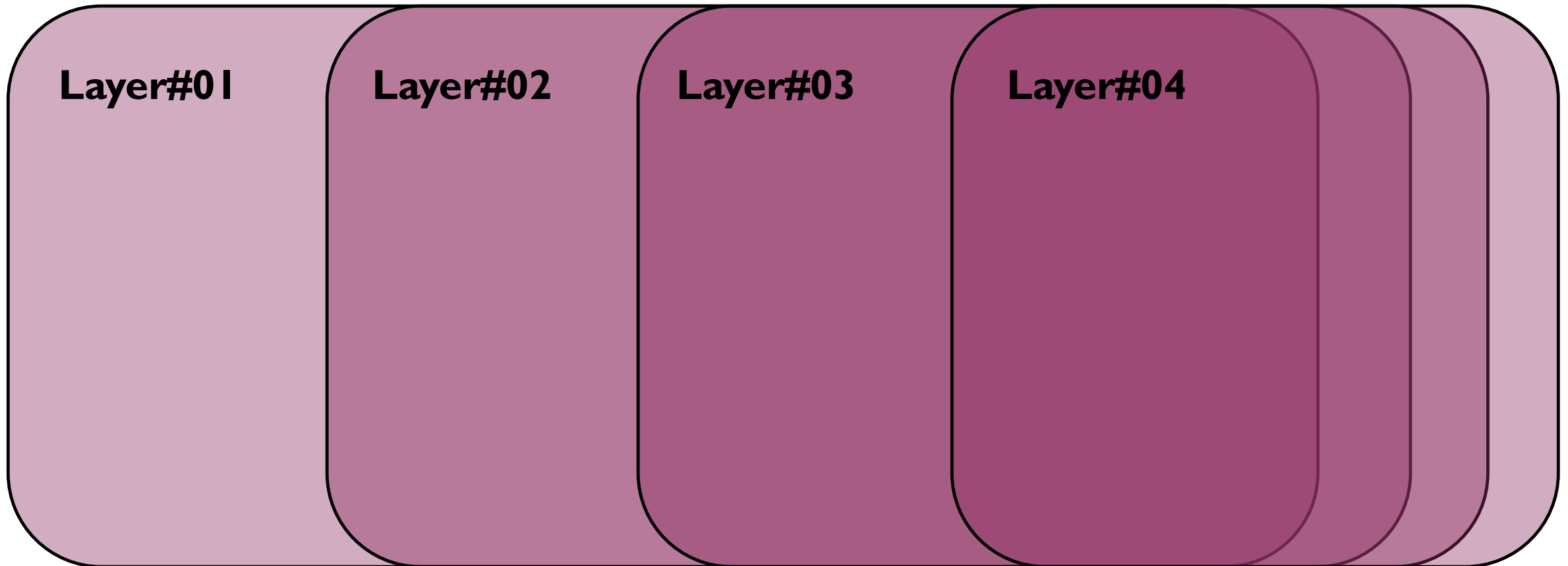
SUPPORT TO THE VULNERABLE HHS

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1. The most poor and vulnerable HHs (TDH Beneficiaries) were selected through the community consultation meeting and based on the set criteria of vulnerability primarily through the Reconstruction Committees.
 2. Participation of most poor and vulnerable people is ensured in RAP meetings through the mobilization of Social Mobilizers and Awas Nirman Saathis (ANSs)
 3. To support the most poor and vulnerable HHs, community consultation meetings were organized and people were invoked to support the most-at-risk HHs by facilitating material (stones & timbers) support or, in-kind human resource (labor) support, so that, their houses are completed. Even in some cases, project team contributed voluntary labor for 1 to 2 days to support those most poor and vulnerable HHs.
 4. ANS support and MTC campaign were more focused on the most poor and vulnerable communities by creating awareness on earthquake resistant features of cost-appropriate technology affordable to them

Cost of implementation per HH per month= 52 € (~ 60 \$)

SELECTION OF HOUSE-OWNERS FOR SUPPORT UNDER DIFFERENT INITIATIVES [TDH & RECONSTRUCTION GRANT]

Selection process of House-owners [Multiple layers of vulnerability]



SELECTION OF HOUSE-OWNERS FOR SUPPORT UNDER DIFFERENT INITIATIVES [TDH & RECONSTRUCTION GRANT]

Selection of Project Participants for TDH

Total 65 [F-38, M-27]

LAYERS of Vulnerability

Social Vulnerability:

Dalit.....	9	} 65
Janajati.....	35	
BCT.....	21	

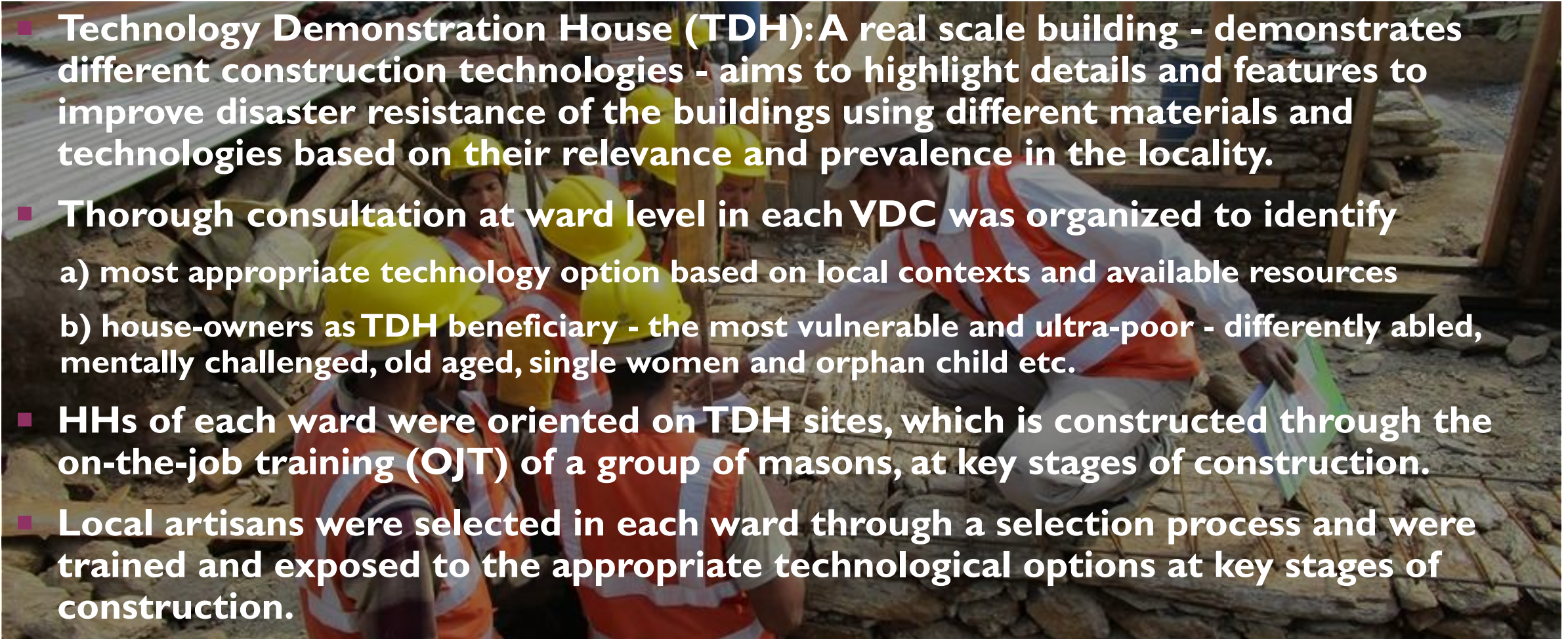
Other Layers of Vulnerability

Single women.....	33	} 65
Physically and mentally challenged....	15	
Orphan.....	1	
Above 60 years age.....	19	

B. APPROACH & IMPLEMENTATION STRATEGY

OJT & CONSTRUCTION OF **TDH** & **KEY STAGE ORIENTATION**

- **Technology Demonstration House (TDH):** A real scale building - demonstrates different construction technologies - aims to highlight details and features to improve disaster resistance of the buildings using different materials and technologies based on their relevance and prevalence in the locality.
- **Thorough consultation at ward level in each VDC** was organized to identify
 - a) most appropriate technology option based on local contexts and available resources
 - b) house-owners as TDH beneficiary - the most vulnerable and ultra-poor - differently abled, mentally challenged, old aged, single women and orphan child etc.
- **HHs of each ward** were oriented on TDH sites, which is constructed through the on-the-job training (OJT) of a group of masons, at key stages of construction.
- **Local artisans** were selected in each ward through a selection process and were trained and exposed to the appropriate technological options at key stages of construction.



B. APPROACH & IMPLEMENTATION STRATEGY

OJT & CONSTRUCTION OF TDH [RESULTS]



Orientation at key stages of construction for house-owners and local masons for replication of safer housing technologies



On-site technical orientation of local artisans by engineer and ANS on safer construction technologies

605 local artisans trained through on-the-job training (OJT) by constructing **Technology Demonstration Houses (TDHs)** for **65 most vulnerable HHs**

B. APPROACH & IMPLEMENTATION STRATEGY

OJT & CONSTRUCTION OF TDH [RESULTS]



Containment Reinforcement System as a earthquake resistant system for the vernacular/ traditional **Stone Masonry in Mud mortar (SMM)**, easy to learn, easy to transport the raw material & cost-effective.

*It was approved by Government of Nepal. **More than 100 Houses have been built** adapting CR System.*

B. APPROACH & IMPLEMENTATION STRATEGY

AWAS NIRMAN SAATHI (ANS)

- **A local trained mason with good experience in building construction - is a friend for the house-owners, who provided guidance on safer reconstruction and on whom they could always rely on and seek for appropriate advice.**
- **Through door-to-door visit - ensured a house-owner incorporates disaster resistant construction features based on NRA guidelines and facilitated for inspection and release of tranche**
- **Supported in promotion of cost-effective technologies and designs - by providing direct supervision during construction and facilitated access to construction materials and masons**



B. APPROACH & IMPLEMENTATION STRATEGY

AWAS NIRMAN SAATHI [RESULTS]



ANS providing technical support to house-owner & construction artisans



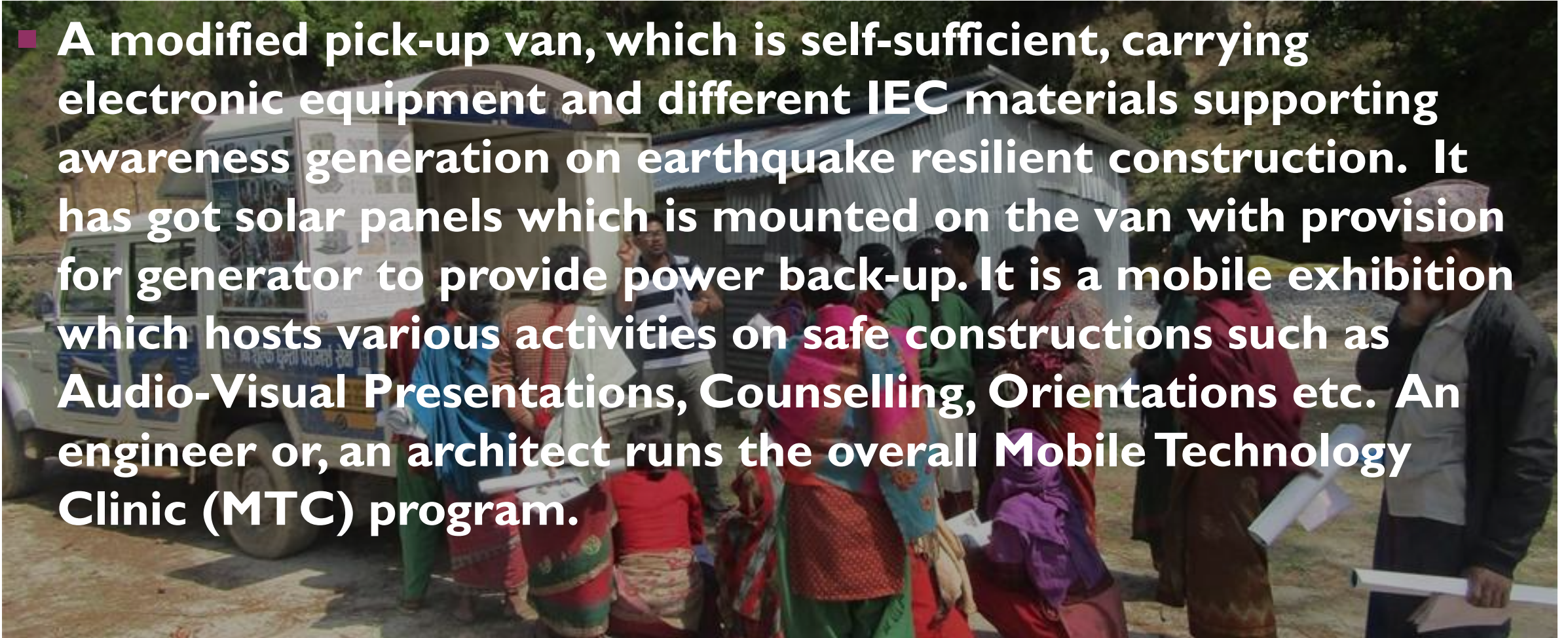
The house-owner and the ANS after satisfactory reconstruction of the house

752 HHs completed construction under direct supervision of ANSs

B. APPROACH & IMPLEMENTATION STRATEGY

MOBILE TECHNOLOGY CLINIC (MTC)

- **A modified pick-up van, which is self-sufficient, carrying electronic equipment and different IEC materials supporting awareness generation on earthquake resilient construction. It has got solar panels which is mounted on the van with provision for generator to provide power back-up. It is a mobile exhibition which hosts various activities on safe constructions such as Audio-Visual Presentations, Counselling, Orientations etc. An engineer or, an architect runs the overall Mobile Technology Clinic (MTC) program.**



B. APPROACH & IMPLEMENTATION STRATEGY

MOBILE TECHNOLOGY CLINIC [RESULTS]

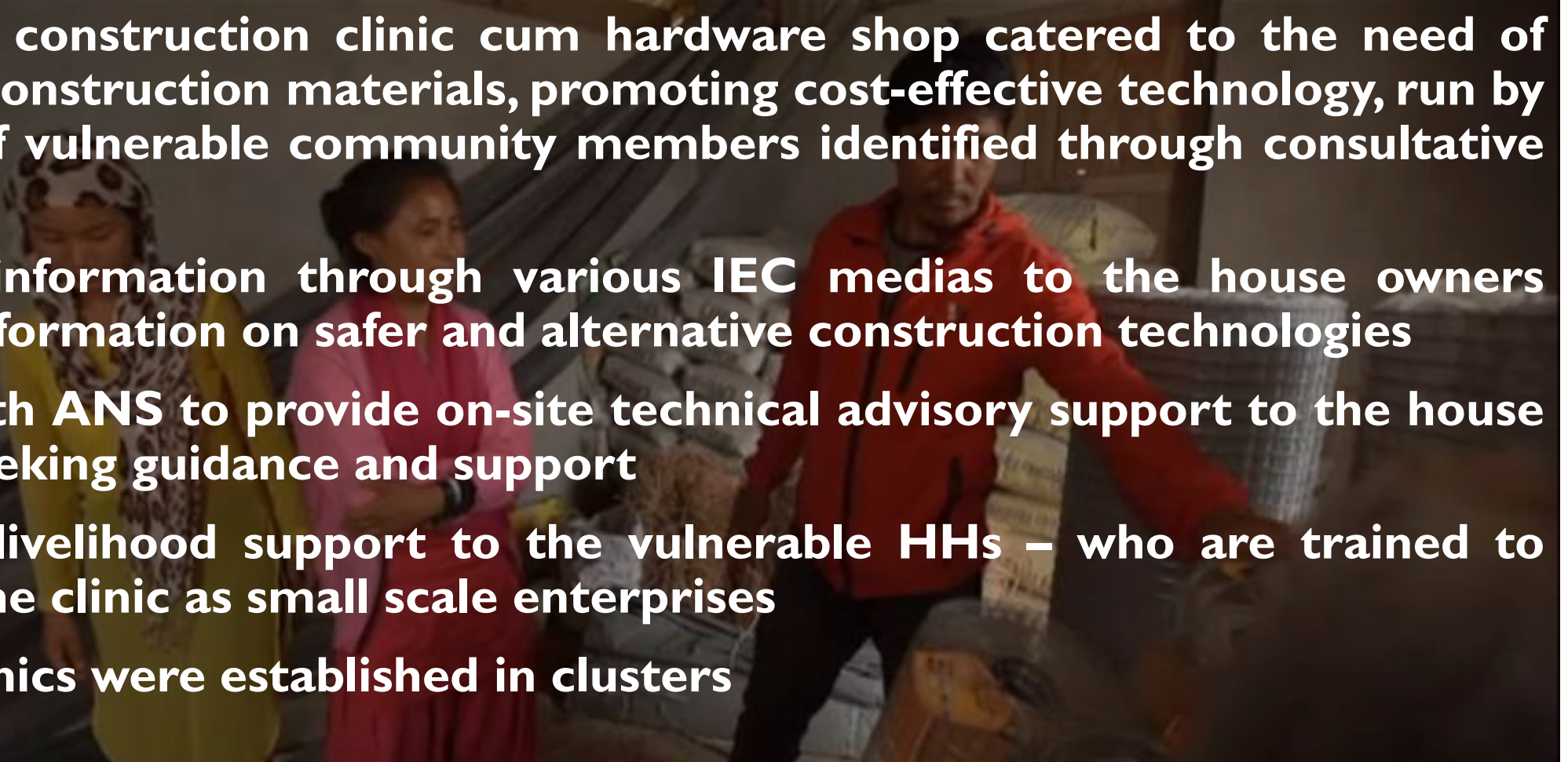


Mobile Technology Clinic (MTC) initiative reached out to **13,765** HHs in Sindhupalchok & Dolakha Districts

B. APPROACH & IMPLEMENTATION STRATEGY

SAFER CONST^N CLINIC CUM HARDWARE SHOP *[RESULTS]*

- The safer construction clinic cum hardware shop catered to the need of required construction materials, promoting cost-effective technology, run by a group of vulnerable community members identified through consultative process.
- Provides information through various IEC medias to the house owners seeking information on safer and alternative construction technologies
- Linked with ANS to provide on-site technical advisory support to the house owners seeking guidance and support
- Provided livelihood support to the vulnerable HHs – who are trained to operate the clinic as small scale enterprises
- 4 such Clinics were established in clusters



B. APPROACH & IMPLEMENTATION STRATEGY

RECONSTRUCTION GRANT

- The most vulnerable households, whose reconstruction process halted at critical stage of construction and needed support to reach the stage to be eligible for the government's NRA grants, were selected for this initiative.
- Selections were based on the set criteria: (a) Vulnerable and marginalized households, (b) enlisted in the government grantee list and do not have regular source of income, (c) unable to reconstruct on their own, (d) not receiving any external support, (e) not having easy access to reconstruction materials
- ANS provided technical support and direct supervision to build safer disaster resilient houses during construction, to the house-owners and masons.
- House-owners were linked with Safer Construction Clinic cum hardware shops to receive in-kind construction material support – further supporting the clinic and linking households to procure material at a reasonable price

B. APPROACH & IMPLEMENTATION STRATEGY

RECONSTRUCTION GRANT *[RESULTS]*



Reconstruction Grant (in-kind material support worth NRs 50k), enabled **100 vulnerable HHs** who couldn't start reconstruction or, reconstruction halted mid-way due to shortage of money or, material

ISSUES/ CHALLENGES

- Community prepared the Reconstruction Action Plan (RAP) by Reconstruction Committees, but there were challenges regarding finding **appropriate solution measures for its implementation**, the process need to be institutionalized at the local level
- House-owners faced challenges for **tranche-release**, for vulnerable HHs it's even more challenging
- In some cases due to higher aspiration house-owners spent more on reconstruction by **borrowing money locally at a high interest rate (24% - 36%)**
- **Continuation of trained artisans in reconstruction**, a challenge because of higher-aspiration
- **Female trainees were less accepted** as capable artisans due to socio-political constraint
- **“Safer Construction Clinic cum Hardware Shop” operated by the most poor & vulnerable groups** selected through the consultative meeting at the local level. Though they received skill enhancement and entrepreneurship training, **need handholding for an extended period of time.**
- **Technical facilitation** provided to the house-owner **differs from entity to entity (person or, organization)**. Common understanding b/w organizations on technical aspects is essential.

SUMMARY OF ACHIEVEMENTS

- **10 VDCs** undertook reconstruction in accordance with the **Reconstruction Action Plan (RAP)** through an inclusive process
- **605 local artisans trained** through on-the-job training (OJT) by **constructing Technology Demonstration Houses (TDHs)** for **65 most vulnerable HHs**
- **752 HHs** incorporated disaster resistant construction features until stage of grant disbursement on the advice from the **Awam Nirman Saathis (ANSs)** and **Mobile Technology Clinics (MTCs)**
- **4 Safer Construction Clinics cum Hardware Shops** established to support the communities to build back better through supply of affordable construction materials
- **100 most-at-risk house-owners received reconstruction grant** (in-kind construction material support worth NRs. 50,000) to build safer disaster resilient houses. They were all technically supported as well.
- **Revolving Fund**, have been established in number of GPs in Gorkha District
Aarughat GP: 10 lakh NRs **Dharche GP:** 15 lakh NRs **Sahid Lakhan GP:** 18 lakh NRs **Gandaki GP:** ~10 Lakh NRs



THANK YOU...!